

AI Resume Screening System

Project Overview

The AI Resume Screening System is an intelligent web application that automates the resume screening process using Artificial Intelligence. It analyzes resumes against a given Job Description (JD), calculates an ATS compatibility score, identifies missing skills, and provides recommendations to improve the resume.

The application helps recruiters shortlist candidates efficiently while also assisting job seekers in optimizing their resumes for Applicant Tracking Systems (ATS).

Features

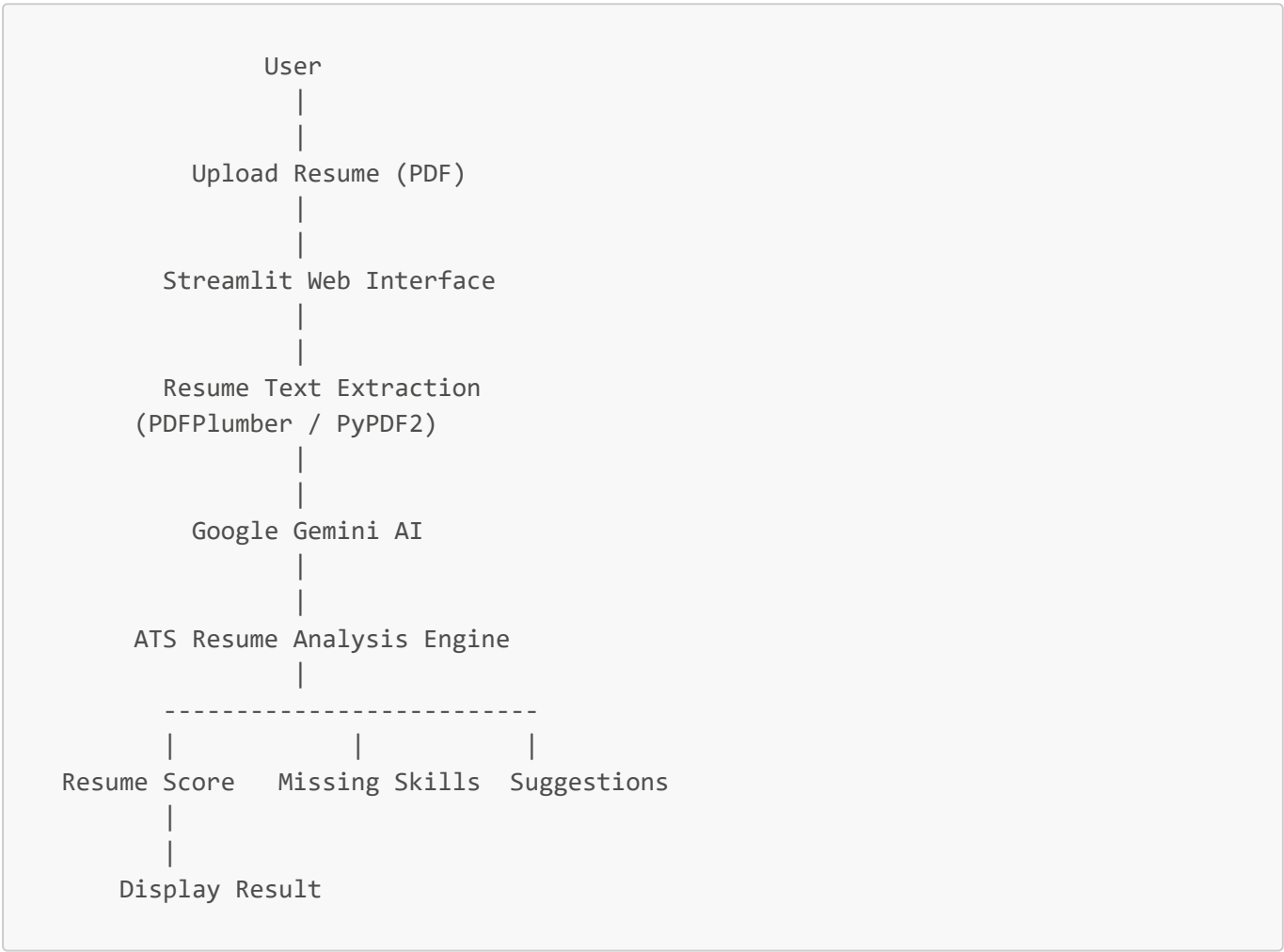
- Upload Resume in PDF format
 - Upload Job Description
 - ATS Resume Score
 - Resume Skill Analysis
 - Missing Skills Detection
 - Resume Summary
 - AI-Based Recommendations
 - Keyword Matching
 - Recruiter-Friendly Interface
 - Fast Resume Processing
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Technologies Used

Technology	Purpose
Python	Backend Development
Streamlit	Web Application
Google Gemini AI	Resume Analysis
PDFPlumber	PDF Text Extraction
PyPDF2	PDF Processing
spaCy	Natural Language Processing
Sentence Transformers	Semantic Similarity
Scikit-Learn	Machine Learning
Pandas	Data Processing
FastAPI	API Development

Technology	Purpose
Uvicorn	API Server
Docker	Containerization
AWS EC2	Cloud Deployment
Git & GitHub	Version Control

Project Architecture



Project Structure

```
AI-Resume-Screening/
├── app.py
├── requirements.txt
├── README.md
├── Dockerfile
└── .env
```

```
|— uploads/  
|— static/  
|— templates/  
|— models/  
|— utils/
```

Implementation Steps

Step 1: Clone Repository

```
git clone https://github.com/yourusername/AI-Resume-Screening.git  
  
cd AI-Resume-Screening
```

Step 2: Create Virtual Environment

```
python -m venv venv
```

Activate Environment

Windows

```
venv\Scripts\activate
```

Linux

```
source venv/bin/activate
```

Step 3: Install Dependencies

```
pip install -r requirements.txt
```

Step 4: Configure Environment Variables

Create a `.env` file.

```
GOOGLE_API_KEY=YOUR_API_KEY
```

Step 5: Run Application

```
streamlit run app.py
```

```
ubuntu@ip-172-31-92-215:~$ cd /mlstorage/ATS_Resume
ubuntu@ip-172-31-92-215:/mlstorage/ATS_Resume$ source .venv/bin/activate
-bash: .venv/bin/activate: No such file or directory
ubuntu@ip-172-31-92-215:/mlstorage/ATS_Resume$ source venv/bin/activate
(venv) ubuntu@ip-172-31-92-215:/mlstorage/ATS_Resume$ streamlit run app.py

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

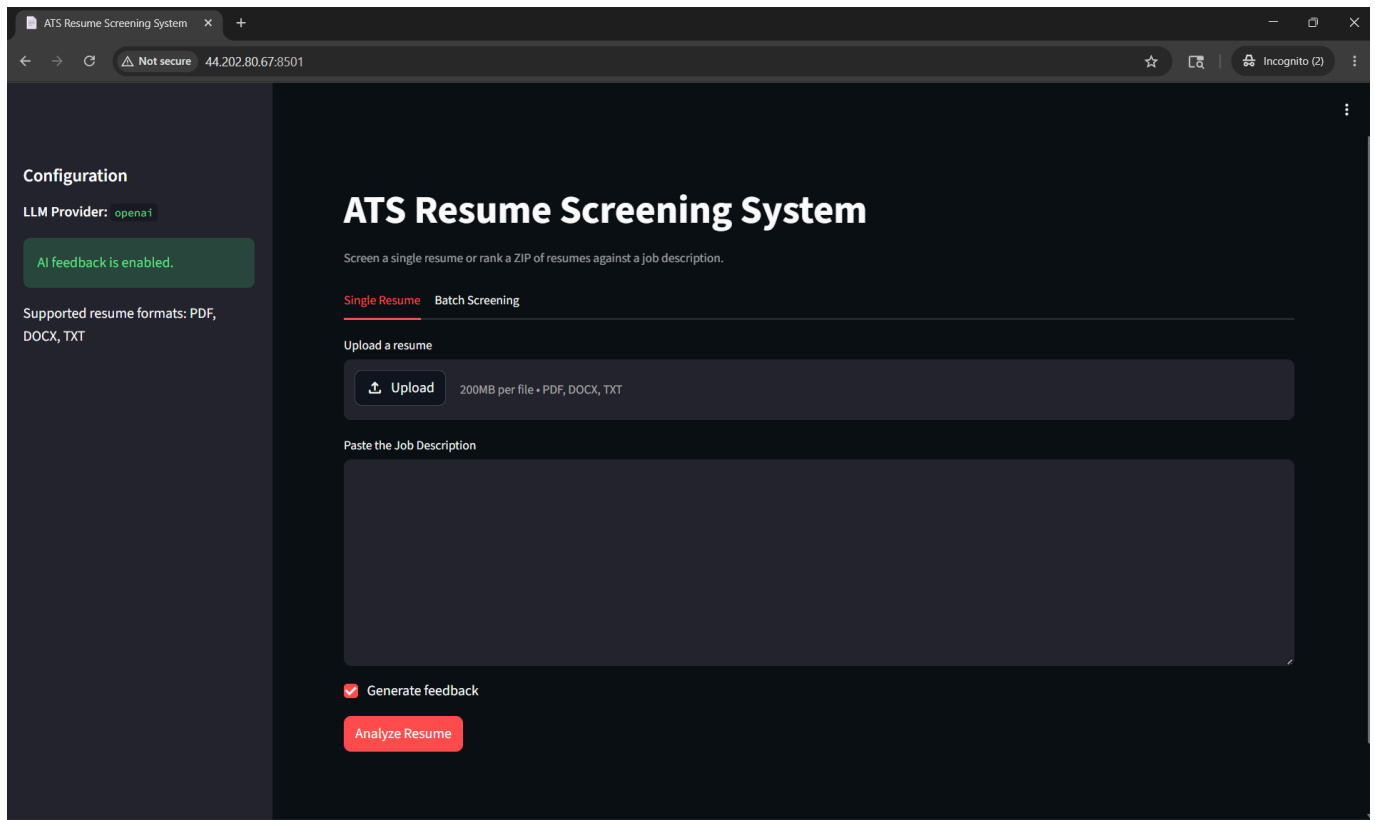
2026-07-25 13:22:47.217 Uvicorn server started on 0.0.0.0:8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://172.31.92.215:8501
External URL: http://34.201.51.38:8501

Invalid HTTP request received.
Invalid HTTP request received.
Invalid HTTP request received.
Invalid HTTP request received.
Invalid HTTP request received.
Invalid HTTP request received.
Invalid HTTP request received.
Invalid HTTP request received.
^C Stopping...
(venv) ubuntu@ip-172-31-92-215:/mlstorage/ATS_Resume$ streamlit run app.py
```

Screenshot



Step 6: Upload Resume

Upload resume in PDF format.

Screenshot

![Upload Resume](images/pdf.png)

Step 7: Upload Job Description

Upload or paste Job Description.

Step 8: AI Resume Analysis

Google Gemini AI processes the resume and compares it with the Job Description.

The system performs:

- Resume Parsing
- Skill Extraction
- Keyword Matching

- ATS Score Calculation
- Missing Skill Detection
- Resume Recommendations

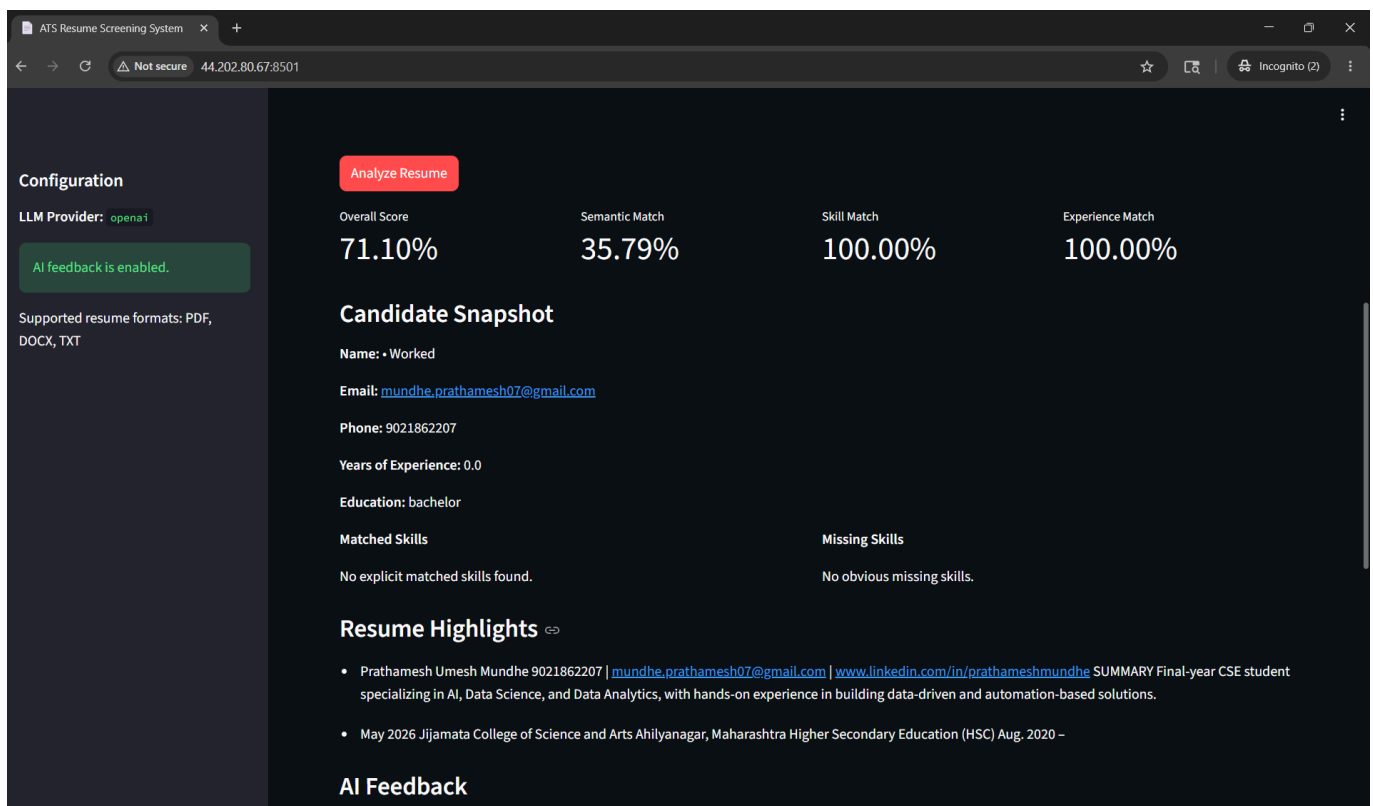
Step 9: ATS Resume Score

Displays ATS Compatibility Score.

Example:

```
```text
ATS Score : 87%
```

## Screenshot



## Step 10: Missing Skills

Shows skills missing from the resume.

Example:

- Docker
- Kubernetes
- AWS
- Terraform

## Screenshot

# Step 11: Resume Improvement Suggestions

Provides AI-generated suggestions to improve the resume.

 Screenshot

### Candidate Snapshot

**Name:** • Worked

**Email:** [mundhe.prathamesh07@gmail.com](mailto:mundhe.prathamesh07@gmail.com)

**Phone:** 9021862207

**Years of Experience:** 0.0

**Education:** bachelor

**Matched Skills**

- data science

**Missing Skills**

No obvious missing skills.

### Resume Highlights

- Prathamesh Umesh Mundhe 9021862207 | [mundhe.prathamesh07@gmail.com](mailto:mundhe.prathamesh07@gmail.com) | [www.linkedin.com/in/prathameshmundhe](https://www.linkedin.com/in/prathameshmundhe) SUMMARY Final-year CSE student specializing in AI, Data Science, and Data Analytics, with hands-on experience in building data-driven and automation-based solutions.
- May 2026 Jijamata College of Science and Arts Ahilyanagar, Maharashtra Higher Secondary Education (HSC) Aug. 2020 –

### AI Feedback

- Worked scored 80.25%. This evaluation looks beyond raw keyword overlap and focuses on how clearly the resume proves fit for the target role.

**ATS Diagnosis**

The resume appears strongest when it directly overlaps with the job's required skills and terminology. The biggest score improvements will usually come from clearer keyword coverage, stronger quantified achievements, and making role relevance obvious in the first half of the document.

## Workflow



## Screenshots

Step	Screenshot
Clone Repository	image/clone.png
Install Requirements	image/install.png
Configure .env	image/env.png
Run Application	image/run.png
Upload Resume	image/upload.png
Upload Job Description	image/jd.png
Resume Analysis	image/analysis.png
ATS Score	image/score.png
Missing Skills	image/missing.png
Suggestions	image/suggestion.png

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## Project Output

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- Resume uploaded successfully
- Job Description analyzed
- ATS Compatibility Score generated
- Missing Skills identified
- AI-generated Resume Suggestions displayed
- Resume evaluated using Google Gemini AI

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## Learning Outcomes

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Through this project I learned:

- Artificial Intelligence
- Natural Language Processing
- Resume Parsing
- ATS Scoring
- Google Gemini AI Integration
- Streamlit Development
- FastAPI Development
- Docker
- AWS Cloud Deployment
- Python Backend Development

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## Future Enhancements

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- Multi-Resume Screening
  - Recruiter Dashboard
  - Resume Ranking
  - Candidate Shortlisting
  - Email Notification
  - Authentication & Login
  - Database Integration
  - Analytics Dashboard
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## Author

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**Onkar Hol**

Cloud & AI Engineer

B.Tech Artificial Intelligence & Machine Learning

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